ZHANG XUETING

Hong Kong Baptist University
KLN, Hong Kong, China
Email: 23481919@life.hkbu.edu.hk



Summary

Ph.D. candidate in Chinese medicine with expertise in polysaccharide structure, purity assessment, and analytical chemistry. Experienced in HPGPC-CAD, methylation analysis, 1D/2D NMR, LC-MS/GC-MS, and enzyme-based diagnostics, with applications in immunomodulation and gut microbiota. First-author publications in Food Hydrocolloids, Carbohydrate Polymers, and Glycoscience & Therapy. Experienced in targeted and untargeted metabolomics of plants and animals, and in the quality evaluation of herbal materials.

EDUCATION

Ph.D., Chinese Medicine

Hong Kong Baptist University, Hong Kong, China
2023.09–present (expected 2027.06)

Master, Chinese Medicine

China Academy of Chinese Medical Sciences, Beijing, China
2018.09–2021.07

Bachelor, Chinese Pharmacy

Heilongjiang University of Chinese Medicine, Harbin, China
2014.09–2018.06

RESEARCH EXPERIENCE

● **HONG KONG BAPTIST UNIVERSITY**, 2023.09–present, PhD Candidate

- Developed enzyme-based diagnostic criteria for polysaccharide purity; identified a hidden rhamnogalacturonan I (RG-I) component in commercial polygalacturonic acid and showed that it drives immunostimulation and hydrogel mechanics (Food Hydrocolloids, 2026).
- Characterized polysaccharide structure by HPGPC-CAD, monosaccharide composition, methylation analysis, 1D/2D NMR, and oligosaccharide profiling.

- Investigated gut-microbiota-mediated degradation of Dendrobium β-1,4-mannan and its anti-colitis activity in mice (Int J Biol Macromol, 2025; co-first author).
- Studied intestinal lymphatic absorption and M-cell targeting of Astragalus polysaccharides (Int J Biol Macromol, 2026; Phytomedicine, 2026).

● **HONG KONG BAPTIST UNIVERSITY**, 2022.04–2023.08, Research Assistant

- Established targeted and untargeted metabolomics workflows for mouse serum, feces, and tissues.
- Supported studies on TCM polysaccharides, gut microbiota, and intestinal immunity.

● **CHINA ACADEMY OF CHINESE MEDICAL SCIENCES**, 2018.09–2021.07, Master's Research

- Conducted plant metabolomics and chemotaxonomy of lotus (*Nelumbo nucifera*) and *Perilla frutescens*.
- Established GC-MS and LC-qTOF-MS methods for fatty acids and flavonoids.
- Built a lotus germplasm nursery and analyzed browning-related metabolites across varieties.

SKILLS

● **Analytical chemistry:** HPGPC-CAD; HPLC/UPLC; GC; GC-MS; LC-MS (qTOF, triple quadrupole); ICP-MS; 1D/2D NMR; methylation analysis; oligosaccharide profiling; enzyme-based purity diagnostics.
● **Metabolomics:** targeted and untargeted metabolomics in plants and animals; sample preparation; method validation; multivariate analysis.
● **Immunology and cell biology:** RAW264.7 macrophage assays; ELISA; Western blot; flow cytometry; H&E staining; Immunofluorescent staining, Confocal, ELISA; Whole-mount staining; Intestinal Loop experiment.
● **Materials characterization:** rheology; SEM; hydrogel preparation.
● **Data analysis:** R; Prism; Origin; SigmaPlot; ImageJ.
● **Languages:** Mandarin; English.

PUBLICATION

1. **Zhang X**[#], Huo C[#], Shan P, et al. Rhamnogalacturonan I in commercial samples of polygalacturonic acid is a functional driver: a study of isolation, structural elucidation, role in immunostimulation and hydrogel mechanics. Food Hydrocolloids. 2026;180:112926. doi:10.1016/j.foodhyd.2026.112926. **(1st author, Q1, IF 13.8)**
2. **Zhang X**[#], Xia J[#], Zhang Q, et al. Enzyme diagnostic criteria for polysaccharide purity: a model study using polymer reference standards. Glycoscience & Therapy. 2026;2:100037. doi:10.1016/j.glycos.2026.100037. **(1st author; new journal)**
3. **Zhang X**[#], Li L[#], Fung H, et al. Critical review of the criterion of polysaccharide purity. Carbohydrate Polymers. 2025;352:123187. doi:10.1016/j.carbpol. 2024.123187. **(1st author, Q1, IF 12.5)**

4. Huo C[#], **Zhang X[#]**, Zhang Q, et al. Vancomycin-sensitive gut microbiota mediate Dendrobii β-1,4-mannan degradation and anti-colitis activity in mice. Int J Biol Macromol. 2025;323(Pt 1):147052. doi:10.1016/j.ijbiomac.2025.147052. **(Co-1st author, Q1, IF 8.5)**
5. Zhao W, **Zhang X**, Shan P, et al. Dynamic distribution of label-free polysaccharides after intestinal lymphatic absorption. Int J Biol Macromol. 2026;375:153477. doi:10.1016/j.ijbiomac.2026.153477. (Co-author, Q1, IF 8.5)
6. Zhang Q, Huo C, Liu M, **Zhang X**, et al. Lymph-absorbed Radix Astragali polysaccharide induces IgA production through TLR4-dependent DC-B axis to alleviate colitis. Phytomedicine. 2026;162:158787. doi:10.1016/j.phymed.2026.158787. (Co-author, Q1, IF 11.3)
7. Li J, Feng M, Chen N, Shan P, Fung H, **Zhang X**, et al. Identification of a commercial xylan as a microfold cell-targeting polysaccharide in vivo and in vitro. Int J Biol Macromol. 2026;376:153501. doi:10.1016/j.ijbiomac.2026.153501. (Co-author, Q1, IF 8.5)
8. Li LF, Shi X, Qi SM, **Zhang XT**, et al. Strategies, techniques and applications for food authentication based on carbohydrates: a review. Carbohydrate Polymers. 2024;344:122533. doi:10.1016/j.carbpol.2024.122533. (Co-author, Q1, IF 12.5)
9. Wong TL, Li LF, Zhang JX, Zhang QW, **Zhang XT**, et al. Oligosaccharide analysis of the backbone structure of the characteristic polysaccharide of Dendrobium officinale. Food Hydrocolloids. 2023;134:108038. doi:10.1016/j.foodhyd.2022.108038. (Co-author, Q1, IF 12.4)
10. Hao C, Yu Y, **Zhang X**, Dong G, Liu Y, Chen S. Nontarget metabolites of rhizomes of edible sacred lotus provide new insights into rhizome browning. Journal of Food Quality. 2022;2022:3943052. doi:10.1155/2022/3943052. (Co-author)
11. **Zhang X**, Feng S, Yang W, Shen Q, Chen S. A method for identifying germplasm that exhibit multiple mutations in the fatty acid synthetic pathway of Perilla frutescens. J Am Oil Chem Soc. 2020;97(11):1191–1201. doi:10.1002/aocs.12417. **(1st author)**
12. Zhang Q[#], **Zhang X[#]**, Liu J, et al. Identification of copy number variation and population analysis of the sacred lotus (Nelumbo nucifera). Biosci Biotechnol Biochem. 2020;84(10):2037–2044. doi:10.1080/09168451.2020.1786351. **(Co-1st author)**
13. Mao C[#], **Zhang X[#]**, Shi J, Chen S. The complete chloroplast genome sequence of *Clematis montana Buch*.-Ham. (Ranunculaceae) and its phylogenetic analysis. Mitochondrial DNA Part B. 2020;5(3):2246–2247. doi:10.1080/23802359.2020.1771225. **(Co-1st author)**
14. Chen S, Liu J, Dong G, **Zhang X**, et al. Flavonoids and caffeoylquinic acids in Chrysanthemum morifolium Ramat flowers: a potentially rich source of bioactive compounds. Food Chemistry. 2021;344:128733. doi:10.1016/j.foodchem.2020.128733. (Co-author, Q1, IF 8.8)
15. Wu L, Sun Y, Yu H, Qian X, **Zhang X**, et al. Rapid determination of saponins in the honey-fried processing of Rhizoma Cimicifugae by near infrared diffuse reflectance spectroscopy. Molecules. 2018;23(7):1617. doi:10.3390/molecules23071617. (Co-author)

*[#] equal contribution. * corresponding author.*

PRESENTATION

- Poster presentation, Rhamnogalacturonan I in commercial polygalacturonic acid is a functional driver, CarboFuture-2026, Nanchang, China, 23–25 October 2026.
- Oral presentation, Research on the structure–activity relationship of TCM polysaccharides based on upgraded enzymatic diagnostic purity criteria, The 5th HTM2026 International Conference on Herbal and Traditional Medicine, Bangkok, Thailand, 22–25 April 2026.
- Oral presentation, Research on the structure–activity relationship of TCM polysaccharides based on upgraded enzymatic diagnostic purity criteria, The 22nd Meeting of the Consortium for Globalization of Chinese Medicine (CGCM), Taiwan, 25–27 July 2026.

AWARDS & ACTIVITY

- Vincent & Lily Woo Fellowship in Memory of Dr Albert Wong, 2025 — Greater Bay Area fellowship; 2 recipients per year; HK$150,000.
- Volunteer, student-led welfare project (Xi'an–Qinghai Lake), 2014; raised funds for impoverished families in Qinghai Province.